



SCHOOL OF COMPUTING
UUM COLLEGE OF ARTS AND SCIENCES
STIJ3134 NETWORK AND SYSTEM SECURITY
QUIZ 3

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GROUP : A

Answer all questions.
Duration: 10 minutes

1. Why computer network needs to be secured? Give two reasons

(2 Marks)

- Protecting Data - It keeps sensitive information safe from unauthorized access and breaches, ensuring privacy for personal and business data.
- Ensuring Operations - It maintains smooth operation of systems and services, preventing disruptions caused by cyber threats and attacks.

2. Explain two example of possible network attack.

(4 Marks)

- Denial-of-Service (DoS) Attack - Causes disruption and downtime, impacting business operations
- Phishing Attack - Can lead to unauthorized access to accounts and sensitive data, posing security risks.

3. What are the differences between link encryption and end-to-end encryption?

(4 Marks)

Link encryption	End-to-end encryption
Protects data between specific points in a network, like routers or switches.	Secures data from sender to recipient, ensuring it stays encrypted throughout its journey.
Safeguards data within a specific network segment, but it's vulnerable once it leaves that segment.	Keeps data encrypted all the way, protecting it from being accessed or altered by anyone other than the sender and recipient.

4. What are the two main concern regarding authentication?

(2 Marks)

- Security - Making sure the authentication process is secure to prevent unauthorized access.
- Usability - Ensuring authentication methods are easy for users to use without sacrificing security.

5. Network security is about a combination of security solution in order to prevent, to detect and to protect the system. List down three possible security solution for an organization as a combination for secure system.

(3 Marks)

- a. Firewalls – help to prevent unauthorized access to the organization's by filtering incoming and outgoing traffic based on predefined security rules.
- b. Intrusion Detection System (IDS) – helps to detect and alert administrators about potential threats or suspicious activities occurring within the network, enabling proactive responses to security incidents.
- c. Endpoint protection (such as antivirus programs, EDR) - helps to protect individual devices (endpoints) from malware and other cybersecurity threats, ensuring comprehensive security across the organization's network.